A corrected classification of rank-two Suzuki geometries

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Abstract

We assemble a complete treatment of the rank-two classification in Leemans's 1998 paper. The solvable parabolic families follow from a common normalizer incidence argument. The missing subfield families are the minimal central covers of a subfield point stabilizer, parametrized by projective lines over the smaller field. Their Frobenius orbits give a corrected classification and exact counting formulas. Integer certificates settle all sixty rows of the numerical table; literal permutation and coset certificates settle the twenty-nine diagrams for Sz(8). The printed statements and their corrections remain distinct. Classical Suzuki structure is used as an attributed premise.

1 Scope and conventions

The original is D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie **39** (1998), 97–120. We retain its equivalence convention: an automorphism of the specified group sends one subgroup pair to the other, with type interchange permitted. Inner equivalence uses inner automorphisms. Thus the classification concerns a geometry together with its acting group. It does not assert that all underlying unmarked incidence graphs are nonisomorphic.

Throughout, $G = \mathrm{Sz}(q)$ with $q = 2^n$, $n \geq 3$ odd. A configuration is an unordered pair of proper subgroups M, L such that $J = M \cap L$ is maximal in each, and at least one of M, L is maximal in G. The corresponding coset geometry is firm, residually connected, flag transitive and residually weakly primitive. The following elementary observation supplies the rank-two reduction used in source Lemma 2.1.

Proposition 1. *Suppose M is maximal in G, $L \not\leq M$, and $J = M \cap L$ is maximal in both parabolics. Then the coset geometry on G/M and G/L, with incidence defined by nonempty intersection, has the four properties above.*

Proof. Every incident pair has a common representative g, and hence is the translate of the base pair (M, L). This proves flag transitivity. The two local degrees are $[M : J]$ and $[L : J]$, both at least two, giving firmness. Its incidence graph is connected exactly when $\langle M, L\rangle = G$: a path is an alternating sequence of cosets and therefore gives a word in the two subgroups; conversely each such word gives a path. Here generation follows from maximality of M. Rank-one residues have no further connectivity condition. The action on G/M is primitive by maximality, and both residue actions are primitive by maximality of J in the corresponding parabolic. These are precisely the residual weak-primitivity conditions in rank two. □

The classical premises are the faithful Suzuki root coordinates, the ovoid action and involution centralizers, the torus normalizers, the subgroup and maximal-subgroup classifications, and $\mathrm{Aut}(G) = G \rtimes \mathrm{Gal}(\mathbb{F}_q/\mathbb{F}_2)$. These are attributed to Suzuki, Tits and Lüneburg, as in the original

Sections 2.2 and 4 and Lemma 6.7. In particular, Tits's ovoid and normal-form theorem in source Lemma 2.3 is retained as a premise, not presented as a new proof. The elementary field assertion in Lemma 2.2 follows because the finite-field automorphisms are the powers of Frobenius: the equation $\sigma^2 : x \mapsto x^2$ is $2i \equiv 1 \pmod{n}$, with exactly one solution for odd n and none for even n.

2 The common arguments and the complete families

We use the detailed proofs in the accompanying notes [2–5]. The first reproves the arithmetic preliminaries and the maximal subgroups of the solvable normalizers, and excludes a point stabilizer as a parabolic. One affine-group argument treats source Lemmas 5.1–5.3. The exclusion of a point stabilizer also gives the original assertion in every rank greater than one of Lemma 6.1; that assertion is not inferred merely from the rank-two case.

The normalizer argument counts embedded overgroups by counting incidences between two subgroup conjugacy classes. Fixing the intersection leaves a normalizer action; inversion and Frobenius then account for type exchange and outer automorphisms. In root coordinates the relevant finite sets are quotients of a Suzuki two-group or its centre. This recovers all solvable maximal-parabolic families and their orbit counts in source Lemmas 6.2–6.6 and 6.8, with the explicit Table 6 correction recorded below.

For the remaining maximal parabolic, write $n = dr$, where r is prime, $d \geq 3$, $K = \mathbb{F}_{2^d}$ and $F = \mathbb{F}_{2^n}$. Let $S = \mathrm{Sz}(2^d)$ and let B_s be its point stabilizer, where $s = 2^d$. In the faithful root coordinates write $Z_T = \{u(0,t) : t \in T\}$. The crucial replacement for the printed subfield row is

$$Q_T = Z_T B_s, \qquad K \leq T \leq F, \qquad \dim_K T = 2. \tag{1}$$

These groups have order $s^3(s-1)$. The source instead gives the larger group of order $qs(s-1)$; that group does not cover B_s when $r \geq 3$.

Theorem 2 (Complete corrected classification)**.** *All configurations are the solvable maximal-parabolic families of [4], together with the following families having maximal parabolic S:*

$J = S \cap L$	L	*condition*
B_s	Q_T	*as in (1)*
$D_{2(s-1)}$	D_{2m}	$m \mid q-1$, $m/(s-1)$ *prime*
$C_{a_\delta(s)} \rtimes C_4$	$C_m \rtimes C_4$	$m \mid a_\epsilon(q)$, $m/a_\delta(s)$ *prime*
$\mathrm{Sz}(2^e)$	$\mathrm{Sz}(2^{er})$	$n = erp$, $r \neq p$ *prime*, $e \geq 3$.

(2)

Here $a_\pm(x) = x \pm \sqrt{2x} + 1$, and ϵ is the unique ambient sign containing $a_\delta(s)$. Central-cover classes are the Frobenius orbits on $\mathbb{P}_K(F/K)$. Each permitted numerical choice in the second or third row gives one class. The last row gives one class for each unordered pair of distinct prime indices. Families appearing under both maximal parabolics are identified once.

Proof. The classical maximal-subgroup list gives a point stabilizer, the three torus normalizers, and the maximal simple subfield groups. The first is excluded by [2], and [4] treats the three normalizers completely. For S, its maximal-subgroup list leaves the intersections in (2).

The key exhaustion argument for the first row is worth isolating. Any solvable overgroup of B_s lies in its uniquely determined ambient point stabilizer $B_q = P \rtimes F^*$. Put $V = L \cap P$. If B_s is maximal in L, the normalizer condition in the two-group V forces its subfield root subgroup U_K to be normal in L. The root-coordinate calculation in [5] gives $N_P(U_K) = Z_F U_K$

and $N_{B_q}(U_K) = Z_F U_K K^*$. Consequently $L \le Z_F B_s$. The subgroup interval above B_s is exactly the lattice of K-subspaces containing K, by [3]. Its minimal elements are (1). Thus these are all covers, not only examples omitted from the printed table. Their equivalence is the stated semilinear line equivalence.

The characteristic odd torus in either of the next two intersections forces its unique ambient torus normalizer. Maximality then makes the torus extension a prime extension. A simple overgroup having that intersection maximal would be the original S, by the uniqueness of a subfield overgroup of prescribed order. For a simple intersection $\mathrm{Sz}(2^e)$, maximality in each parabolic gives two prime relative degrees; uniqueness excludes equal indices. The case $2^e = 2$ is solvable and belongs to the preceding nonsplit row, so it is not counted here. These are the exhaustion arguments proved in full in [5]. Finally, type interchange accounts for the stated identifications. □

3 The exact counting formulas

Let $\omega(m)$ be the number of distinct prime divisors of m. For the first row of (2), define $C(n,d)$ as follows. For $0 \le j < n$, put $h = \gcd(d,j)$ and $g = \gcd(n,j)$, using $\gcd(a,0) = a$. Then

$$C(n,d) = \frac{1}{n}\sum_{j=0}^{n-1} b_j, \qquad b_j = \begin{cases} (2^{h(r-1)}-1)/(2^h-1), & g = rh, \\ \gcd(r, 2^h-1) - 1, & g = h. \end{cases} \tag{3}$$

There are no other cases. The proof in [5] counts fixed projective lines by finite-field Hilbert 90 and averaging; it also covers $r \mid d$. Thus no coprimality of the absolute and relative degrees is assumed.

For composite n, the total number $T(n)$ of configurations is

$$T(n) = 1 + E(n) + \sum_{\substack{r \mid n \text{ prime} \\ d = n/r \ge 3}} \big(C(n,d) + A(n,d)\big), \tag{4}$$

where

$$E(n) = \#\{\{r,p\} : r \ne p \text{ prime},\ rp \mid n,\ n/(rp) \ge 3\},$$
$$A(n,d) = \omega\left(\frac{q-1}{2^d-1}\right) + \sum_{\delta=\pm 1} \omega\left(\frac{t_\delta(q,2^d)}{a_\delta(2^d)}\right).$$

Here $t_\delta(q,s)$ denotes the unique $a_\epsilon(q)$ divisible by $a_\delta(s)$. The constant term is the universal pair $(D_{2(q-1)}, \mathbb{F}_q \rtimes \mathbb{F}_q^*)$.

For prime n, put

$$v = \frac{q-2}{2n}, \quad I = 1 + \frac{(q-2)(q+4)}{4n}, \quad H = 1 + \frac{(q-2)(q+2)}{4n}.$$

Let η_0, η_+, η_- indicate primality of $q-1, a_+(q), a_-(q)$. Then

$$\begin{aligned} T(n) = 1 &+ \eta_0[I + 2v + 1 + H(\omega(a_+(q)) + \omega(a_-(q)))] \\ &+ \sum_{\epsilon=\pm 1} \eta_\epsilon[2v + (v+1)\omega(a_{-\epsilon}(q))] - \eta_+\eta_-(v+1). \end{aligned} \tag{5}$$

Both formulas follow by adding the complete family counts. In (4), the simple subfield pair is counted by an unordered pair of indices; in (5), the last term removes the twice-counted mixed nonsplit family. These are the only overlaps, as proved in [5]. In particular the source Corollary 7.1 is recovered: if n is prime and all three odd factors are composite, then $T(n) = 1$.

4 Numerical and diagram applications

The integer-only criterion in [6] certifies all three odd factors for each intended exponent $e = 1, \ldots, 60$, where $n = 2e + 1$. The result is 180 complete factorizations using 694 recursively certified primes. The exact products, primality witnesses, prime counts and all sixty substitutions in (4)–(5) are retained as readable data and certificates. Of the printed total entries, eighteen exact values and eleven lower bounds are recovered. Thirty exact values require correction, and the lower bound at $e = 39$ is false:

$$T(79) = 19128573095168183144089 < 10^{23}.$$

The minus factor has three distinct prime divisors at $e = 43$ and four at the intended $e = 60$, rather than the printed two and three. The last row is literally labelled 30; its position and parameters specify 60. We record that contextual correction explicitly.

For $q = 8$, [7] acquires the already verified permutation model and subgroup classes. Incidence counting proves exhaustion of the embedded pairs; normalizer actions and type exchange give their equivalence classes. Equivariant label maps certify the actual coset spaces, and literal graph searches certify diameters, girths and local double transitivity. There are exactly 29 outer classes and 75 inner classes, recovering Corollary 8.1. All printed diagram entries are joined to these data, subject to the following explicit corrections:

- In diagrams 25 and 26, residue orders $(12, 4)$ become $(4, 12)$ in the printed parabolic orientation.
- Their inner multiplicities are interchanged: the edge triple $(6, 4, 6)$ has multiplicity one and $(5, 3, 6)$ has multiplicity three.
- Diagrams 27 and 29 are not residually quasiprimitive. The order-eight parabolic admits three different saturated chains to G [8].

The other positive primitivity marks are proved in [7], not inferred from graph appearance. Repeated indistinguishable diagram entries are compared as multisets; no arbitrary unique identification is imposed.

5 Disposition of the source and attribution

The arithmetic and subgroup preliminaries in Lemmas 3.1–3.7 and 5.1–5.3 are recovered by [2]. Lemma 3.2 needs its proper-subfield domain made explicit: $q = s^r$, $s > 2$, $r > 1$. Lemma 6.1 is recovered in all ranks greater than one. The solvable families in Lemmas 6.2–6.6 and 6.8, and Tables 2–7, follow from [4]. In Table 6 the condition that p divide β_q is replaced by $p \mid \alpha_q$, while the hypothesis that β_q be prime remains. Lemma 6.7 and Table 1 are the attributed classical subgroup inputs.

Lemma 6.9 and Table 8 require the complete replacement (1). The corrected Theorem 7.1 and Tables 9–11 are Theorem 2 and (3)–(5), with the local double-transitivity classification of [5]. The

universal pair is locally doubly transitive, as is every central cover. The only additional locally doubly transitive families occur at $q = 8$, with parabolics $(C_5 \rtimes C_4, C_5 \rtimes C_4)$ or $(C_5 \rtimes C_4, C_4 \times C_2)$. Therefore the Section 9 assertion that only the universal family survives for $e > 2$ is false when n is composite. For prime $n > 5$ it is recovered. The quoted numerical examples at $e = 3, 6, 9$ are respectively 2411, 6453727 and 18084183211. The informal assertion that subfield cases have only a small number of geometries supplies no formal bound; the corrected exact totals replace the printed numerical evidence for it.

The source's announced future studies of higher ranks and of $\mathrm{PSL}(2, q)$ are not asserted results of this manuscript. They are not converted into extra obligations or solved-conjecture claims. Every stated classification family, numerical row and rank-two diagram of the fixed original has a disposition above. The complete treatment retains the original refutation of the subfield row and supplies proofs of the corrected classification and all surviving claims.

The contribution of this treatment is the common normalizer and scalar subspace arguments, their complete exhaustion and orbit analysis, and the explicit arithmetic and graph certificates. The source results remain attributed to their original authors; classical Suzuki structure and standard finite-field and group-action results remain borrowed premises. No claim of priority for the standard tools, no Lean verification, and no measured compression ratio is made here. The present assembly is submitted for distinct whole-manuscript review.

References

[1] D. Leemans, *The rank 2 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Beiträge zur Algebra und Geometrie **39** (1998), 97–120.

[2] Human Compression Project, *Maximal subgroups and elementary reductions for Suzuki geometries*, accompanying proof note.

[3] Human Compression Project, *Scalar subspaces in rank-two Suzuki geometries*, accompanying proof note.

[4] Human Compression Project, *Normalizer fibres in rank-two Suzuki geometries*, accompanying proof note.

[5] Human Compression Project, *Subfield parabolics and the corrected Suzuki classification*, accompanying proof note.

[6] Human Compression Project, *Exact arithmetic for the Suzuki classification table*, accompanying proof note.

[7] Human Compression Project, *Finite certificates for the rank-two geometries of* $\mathrm{Sz}(8)$, accompanying proof note.

[8] Human Compression Project, *Two corrections to the diagrams for* $\mathrm{Sz}(8)$, accompanying proof note.